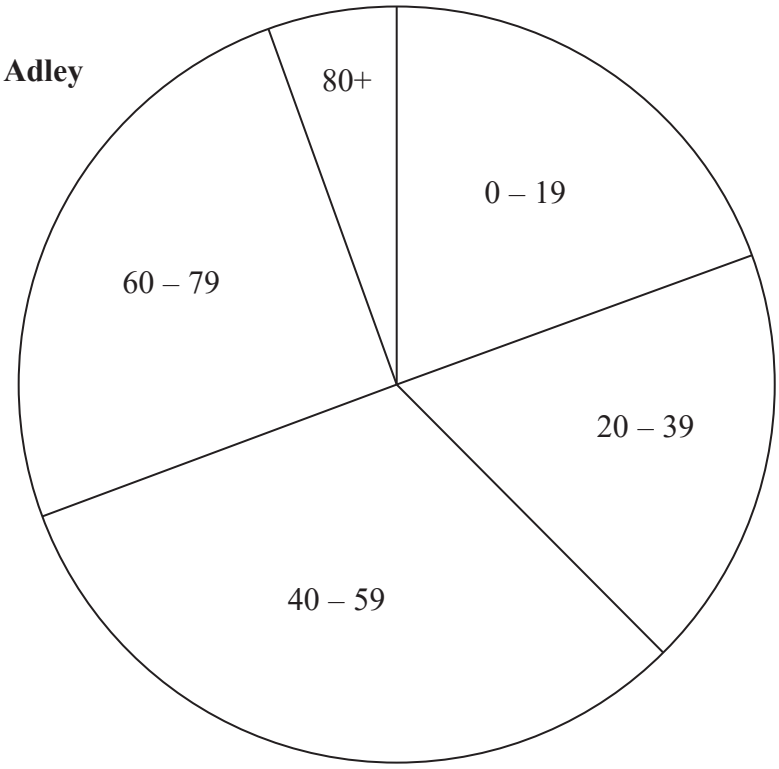
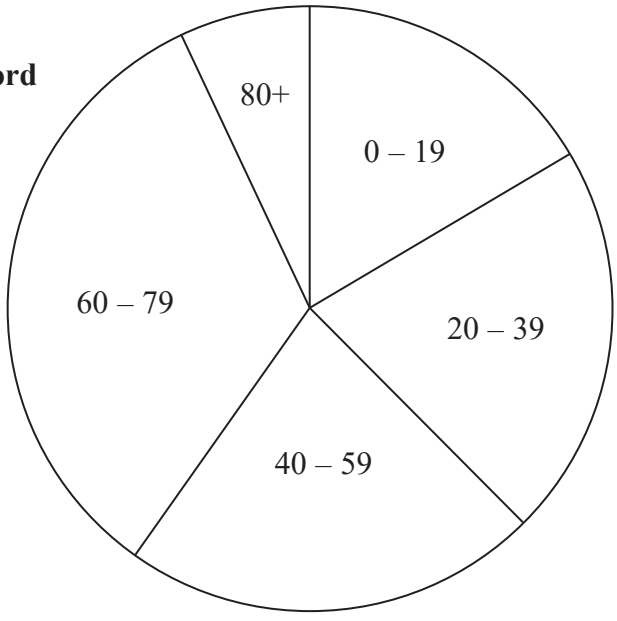
11 The pie charts give information about the ages, in years, of people living in two towns, Adley and Bridford.

Adley



Diagrams accurately drawn

Bridford



At native print size: Adley diameter 10 cm; Bridford 8 cm.

Measure Adley's 0–19 sector: 70°.

1 mark
Measure the radii or sector angle

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

The ratio of the number of people living in Adley to the number of people living in Bridford is given by the ratio of the areas of the pie charts.

What proportion of the total number of people living in these two towns live in Adley **and** are aged 0 – 19?
Give your answer correct to 3 significant figures.

Radii: Adley 5 cm; Bridford 4 cm.

Areas: Adley 25π cm^2; Bridford 16π cm^2.

A	B
25	16

Population ratio A : B = 25 : 16.

Adley is 25/41 of both towns; 70/360 of Adley is aged 0–19.

Required proportion = $\frac{25}{41} \times \frac{70}{360}$

1 mark
Combine the population share and sector fraction

= 175/1476 = 0.11856368...

1 mark
Correct to 3 significant figures

0.119

(Total for Question 11 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA